Indian Disease Data Analysis

Comprehensive analysis of 10,000 patient records to uncover disease patterns, recovery trends, and healthcare insights using Python, SQL, and Power BI.

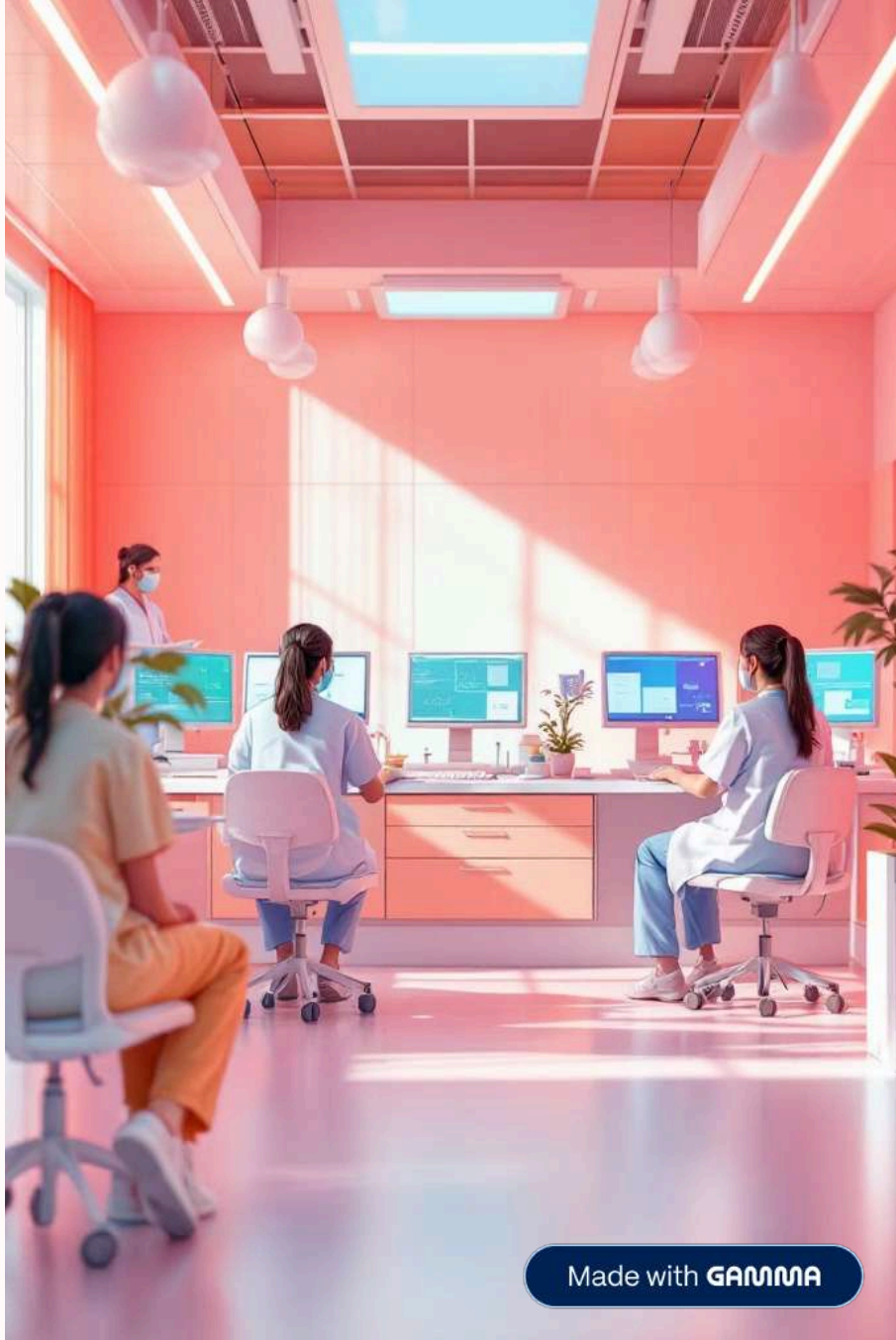
Project Overview

Objective

Clean and process patient data using Python, store refined dataset in MySQL database, and identify relationships between diseases, commodities, recovery time, and lifestyle factors.

Impact

Support better healthcare insights through data-driven analysis of disease patterns, patient health conditions, and recovery trends across India.



Dataset Summary

10K

Patient Records

Comprehensive dataset with detailed health information

12

Data Fields

Demographics, diseases, conditions, and outcomes

100%

Processed

Complete data cleaning and standardization

Patient Demographics

Age group, blood group, BMI classification

Disease Information

Disease name, symptoms, diagnosis date

Health Conditions

Co-morbidity, alcohol use, recovery days

Medical Details

Diagnosis date, recovery time, cause of death

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Data Cleaning Process

01

Data Loading

Import dataset using pandas into DataFrame for analysis

02

Missing Values

Co-morbidity and alcohol use filled with "never"; cause of death as "Unknown"; recovery days imputed using median by disease

03

Standardization

Correct inconsistent values like "19-oct" to "0-20" and "80+" to "80-90"

04

Format Conversion

Convert diagnosis date to datetime format for time-based analysis

05

Database Integration

Connect to MySQL using SQLAlchemy and load cleaned data

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SQL Analysis: Key Metrics

Total Patients

Complete count of registered patients in the dataset

Total Deaths

Number of deaths recorded in dataset

Highest Case Disease

Disease with most reported cases identified

Highest Mortality Rate

Disease with greatest fatality percentage

Average Recovery Time

Calculated by disease category

Gender Distribution

Male vs female patient comparison

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SQL Analysis: Advanced Insights

Monthly Case Trends

Number of cases reported each month based on diagnosis date to identify seasonal patterns

Alcohol vs Recovery

Compare average recovery times between alcohol users and non-users to assess impact

Co-morbidity Risk

Check if patients with co-morbidities have higher death risk compared to those without

Longest Recovery Cases

Identify five patients with longest recovery duration for outlier analysis

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Power BI Dashboard

Interactive visualizations built in Power BI to present healthcare insights clearly and enable stakeholders to explore data trends.

Key Findings

High-Risk Diseases

Diseases with higher case counts or mortality rates require greater attention from healthcare authorities

Co-morbidity Impact

Patients with existing health conditions show higher risk of complications and mortality

Recovery Patterns

Monthly case trends reveal seasonal variations in disease incidence

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Business Recommendations

1 Focus on High-Risk Diseases

Healthcare authorities should prioritize diseases with higher case counts or mortality rates for prevention and treatment planning

2 Early Detection Programs

Promote regular health check-ups and early diagnosis to reduce disease severity and improve recovery rates

3 Lifestyle Awareness Campaigns

Educate people about impact of alcohol consumption and unhealthy habits on disease recovery

4 Monitor Co-morbidity Patients

Closely monitor patients with existing conditions as they face higher complication and mortality risks

5 Resource Allocation

Allocate more resources, staff, and facilities to diseases with longer recovery times or higher case volumes

6 Data-Driven Planning

Regularly analyze healthcare data to identify trends, monitor recovery patterns, and improve decision-making

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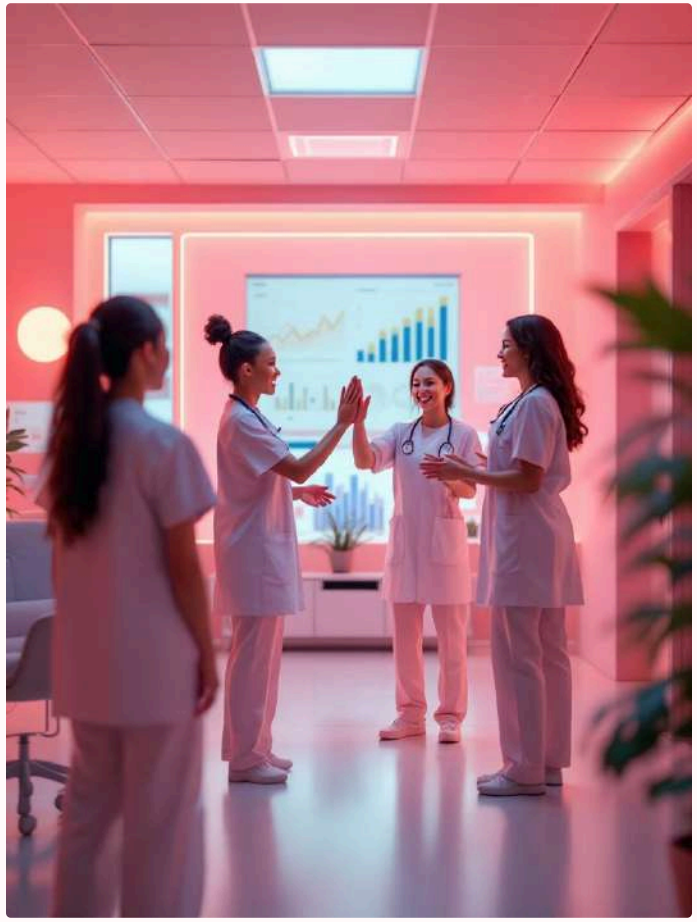
Conclusion

This comprehensive analysis of 10,000 patient records demonstrates how data-driven approaches can transform healthcare decision-making. By combining Python for data cleaning, SQL for analysis, and Power BI for visualization, we uncovered critical insights into disease patterns, recovery trends, and risk factors.

The findings provide actionable recommendations for healthcare authorities to improve patient outcomes through targeted interventions, early detection programs, and better resource allocation.



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